

AI Usage

Operating Systems and Networks Mini Project 1

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Purpose

AI tools were used as development aids during the project. They were used to understand requirements, discuss implementation strategies, debug selected issues, and review code. The final implementation was compiled and tested manually by the student.

AI Tools Used

Tool	Use
ChatGPT	Implementation discussion, debugging, code review, formatting cleanup and testing guidance.
Google Gemini	Planning, debugging and discussion of C-Shell implementation issues.

Main AI-Assisted Tasks

- **C-Shell design:** discussed lexer, parser, command execution, redirection, pipelines, sequential execution and background execution.
- **Job control:** discussed process groups, terminal control, stopped jobs, activities, resume and ping.
- **F1 spy:** discussed using /proc, readlink(), file descriptors and memory mappings.
- **F2 snoop:** discussed ptrace, syscall entry/exit handling, timing, aggregation and sorting.
- **Debugging:** used AI feedback to identify and fix issues such as intrinsic commands inside pipelines and formatting/readability problems.
- **Testing:** AI helped define focused test cases. The resulting programs were compiled and run manually before committing changes.

Example Prompts

- “Give me the implementation approach for the required C-Shell feature and explain the important edge cases.”
- “Fix the issue where intrinsic commands do not work correctly inside pipelines without changing unrelated logic.”
- “Implement snoop using ptrace and summarize syscall counts and execution time.”
- “Review the code formatting: keep useful one-line comments, necessary line breaks and remove unnecessary spacing.”
- “Give me focused tests to verify the implementation before committing it.”

Student Responsibility

AI-generated suggestions were treated as assistance rather than as a substitute for understanding or verification. Code was integrated, adjusted and tested in the project repository. The student is responsible for the submitted implementation.

Relevant Project Evidence

The project history contains incremental commits for the implemented C-Shell parts. The final C-Shell build was compiled with strict warning flags, and features including snoop command tracing and snoop -p process attachment were tested.